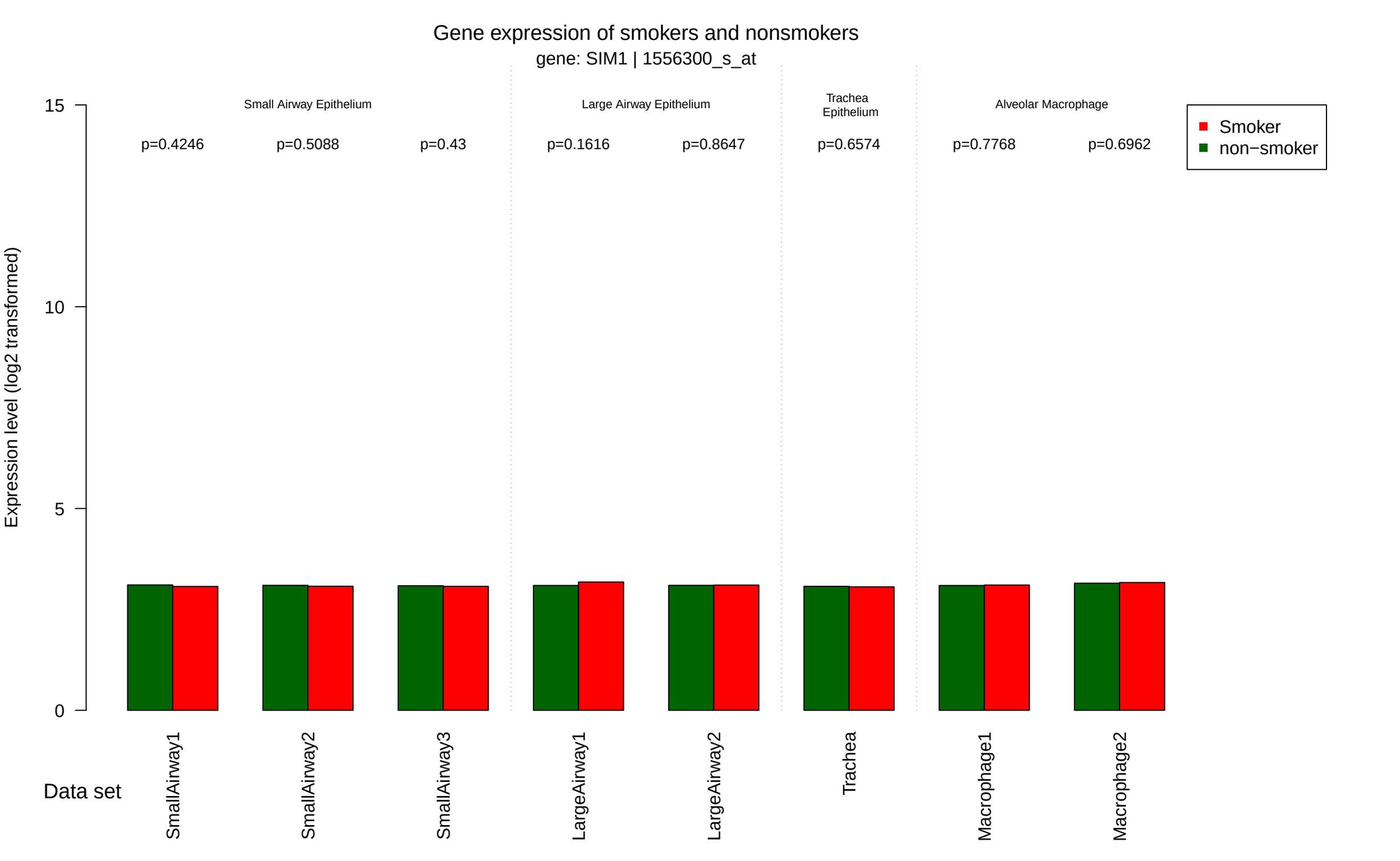
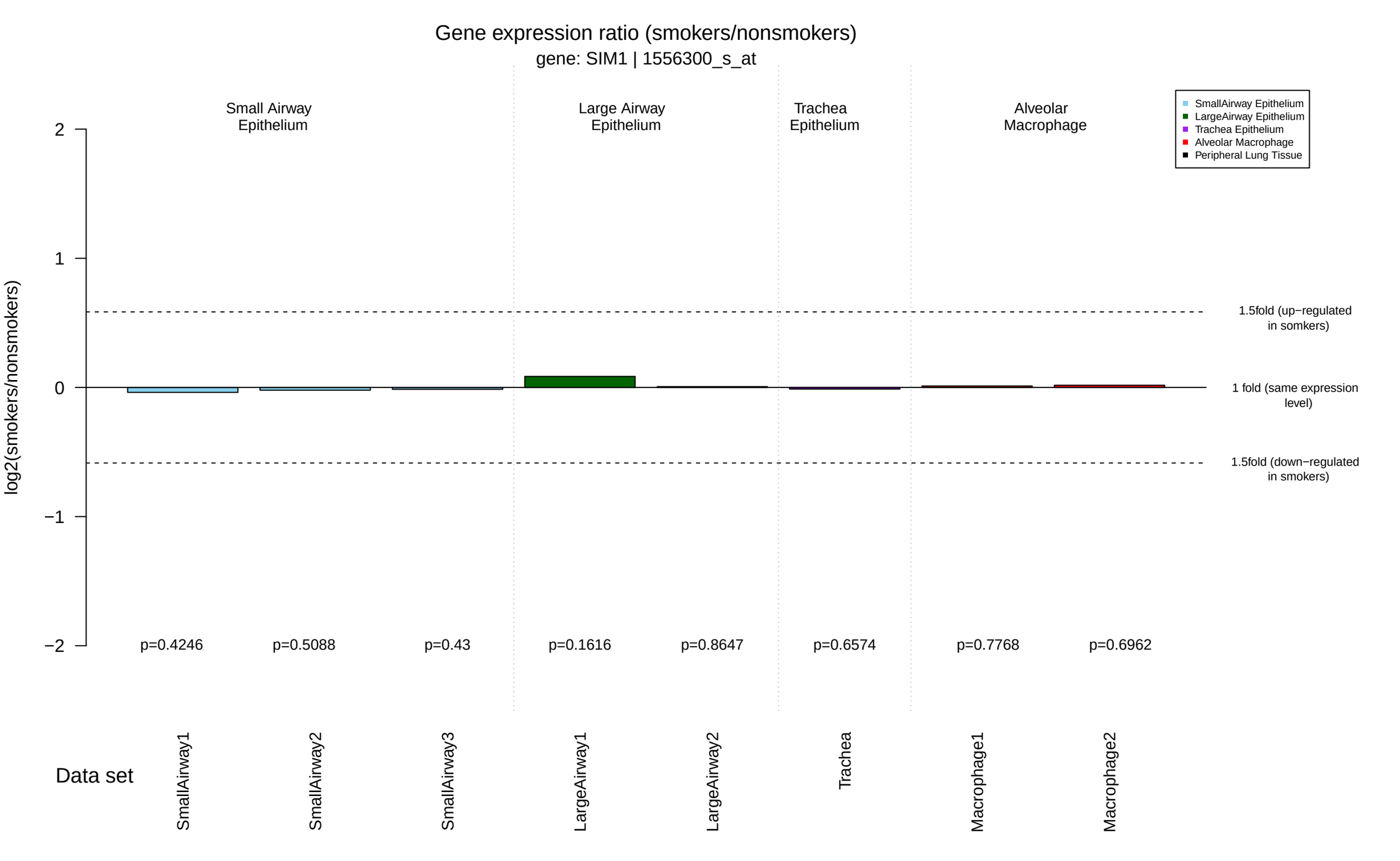
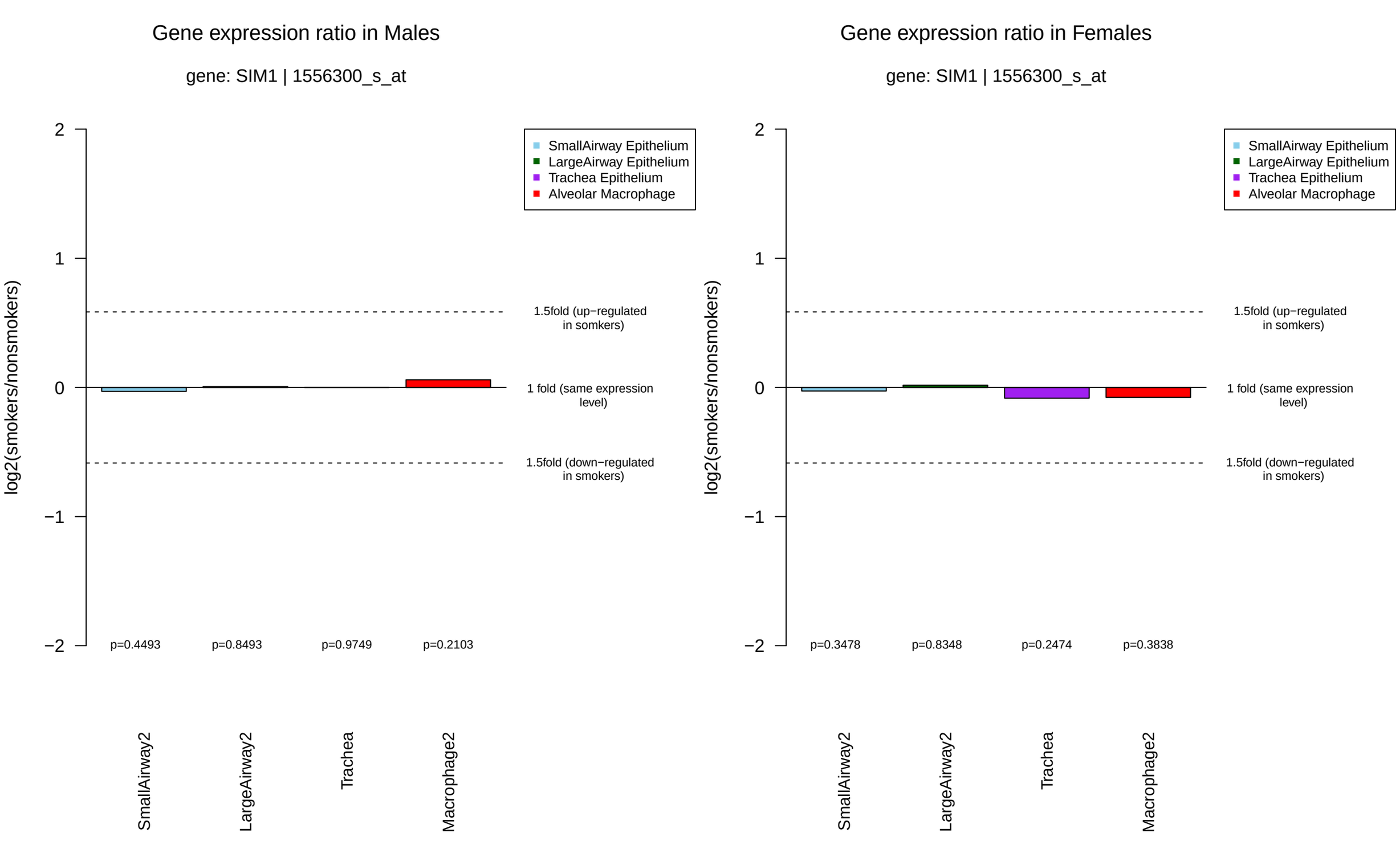
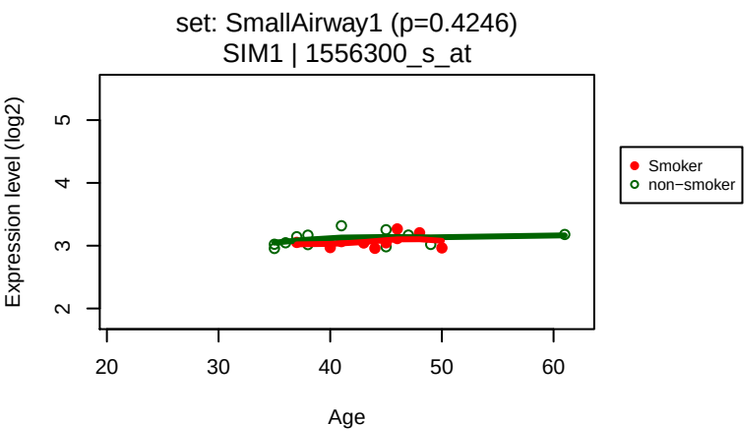
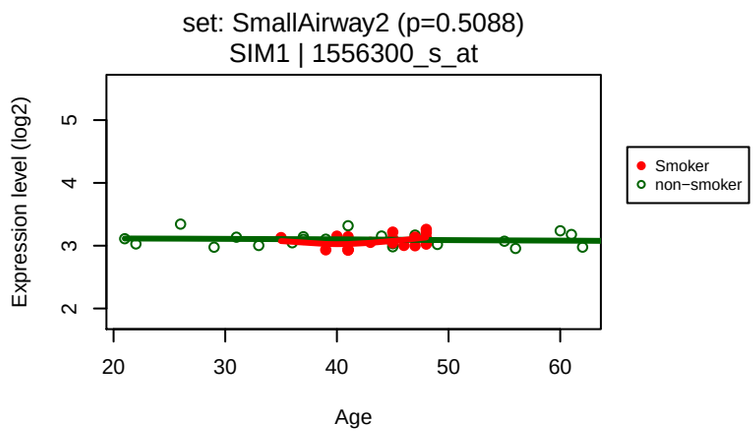
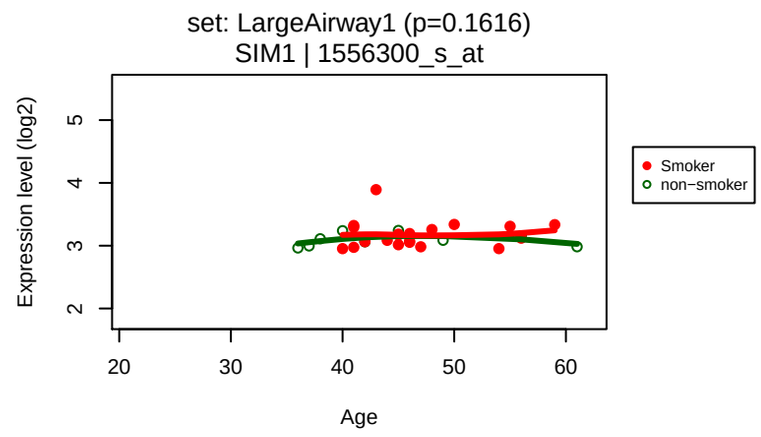

Gene expression of smokers and nonsmokers
gene: SIM1 | 1556300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4246
p=0.5088
p=0.43
p=0.1616
p=0.8647
p=0.6574
p=0.7768
p=0.6962
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

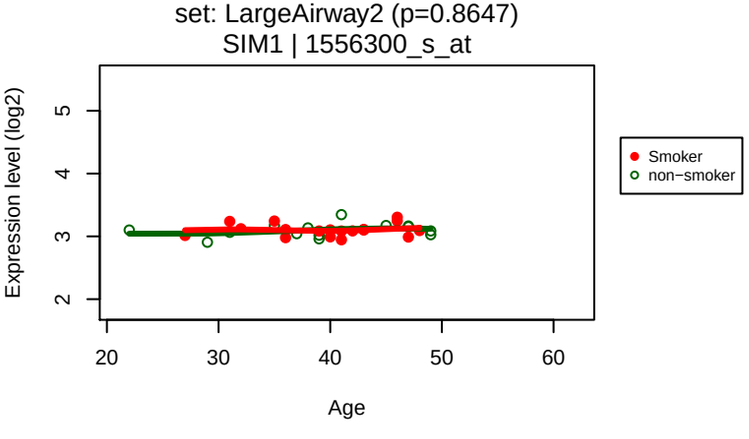

Gene expression ratio (smokers/nonsmokers)
gene: SIM1 | 1556300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4246
p=0.5088
p=0.43
p=0.1616
p=0.8647
p=0.6574
p=0.7768
p=0.6962
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

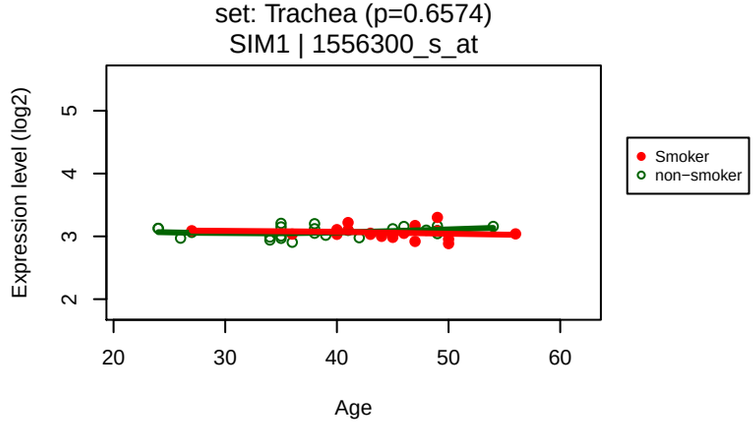

Gene expression ratio in Males
gene: SIM1 | 1556300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4493
p=0.8493
p=0.9749
p=0.2103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SIM1 | 1556300_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3478
p=0.8348
p=0.2474
p=0.3838
SmallAirway2
LargeAirway2
Trachea
Macrophage2

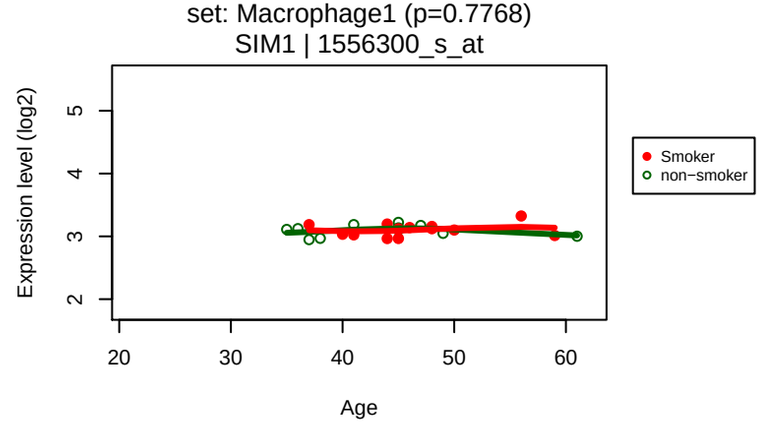

set: SmallAirway1 (p=0.4246)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker

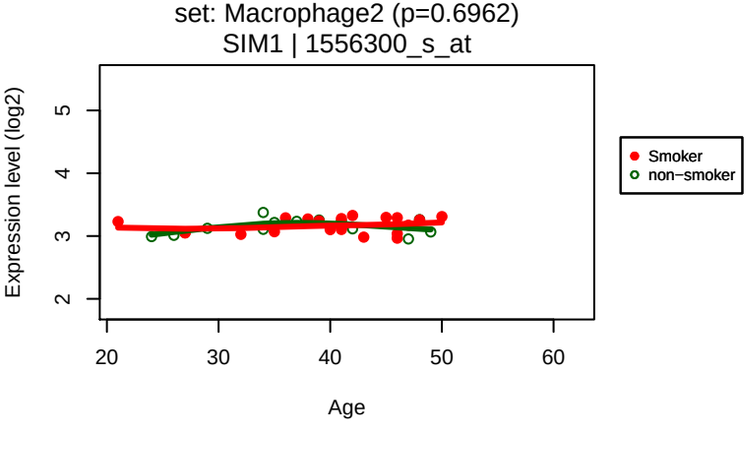

set: SmallAirway2 (p=0.5088)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker

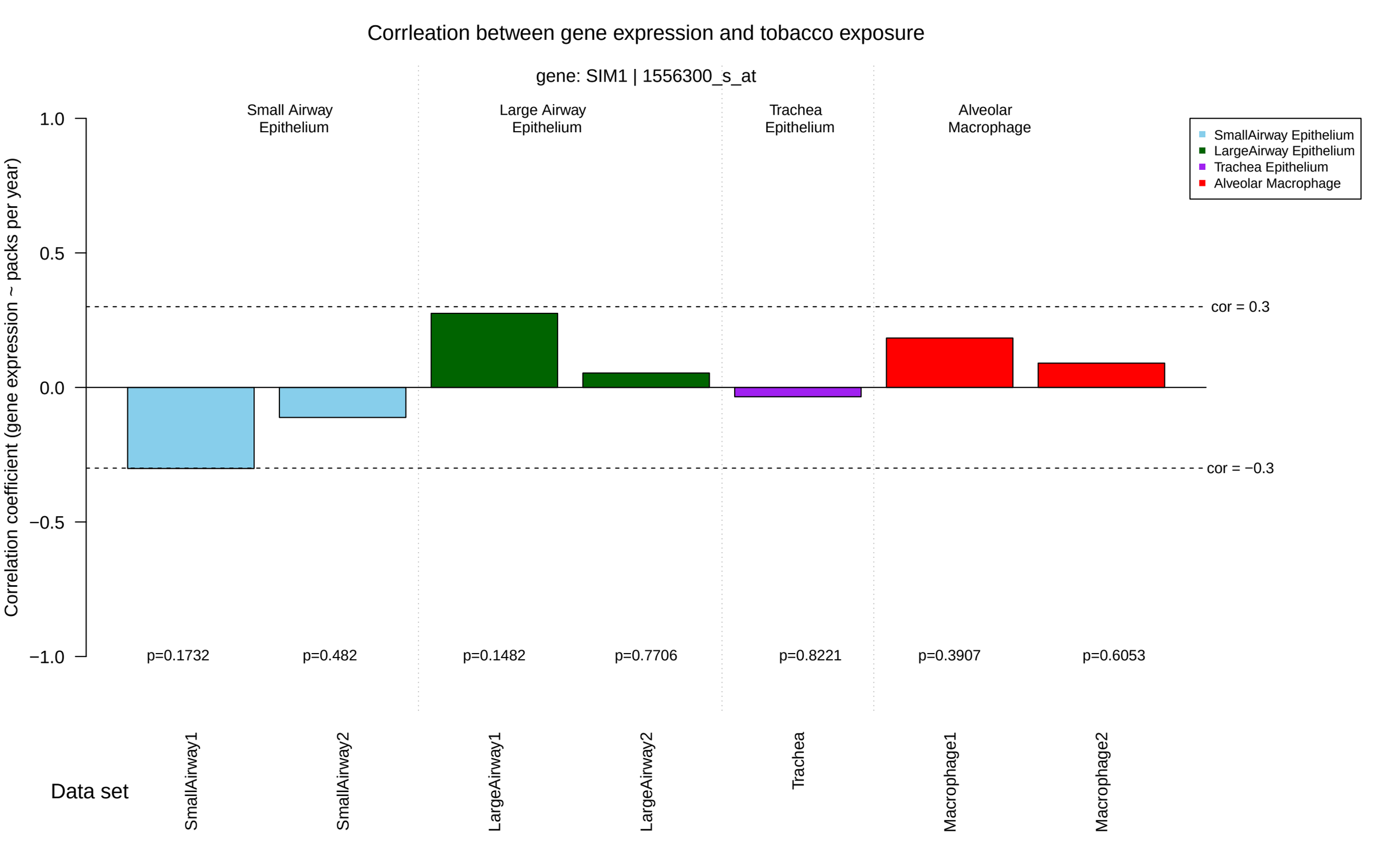

set: LargeAirway1 (p=0.1616)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8647)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6574)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7768)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6962)
SIM1 | 1556300_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SIM1 | 1556300_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1732
p=0.482
p=0.1482
p=0.7706
p=0.8221
p=0.3907
p=0.6053
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2